

Figures

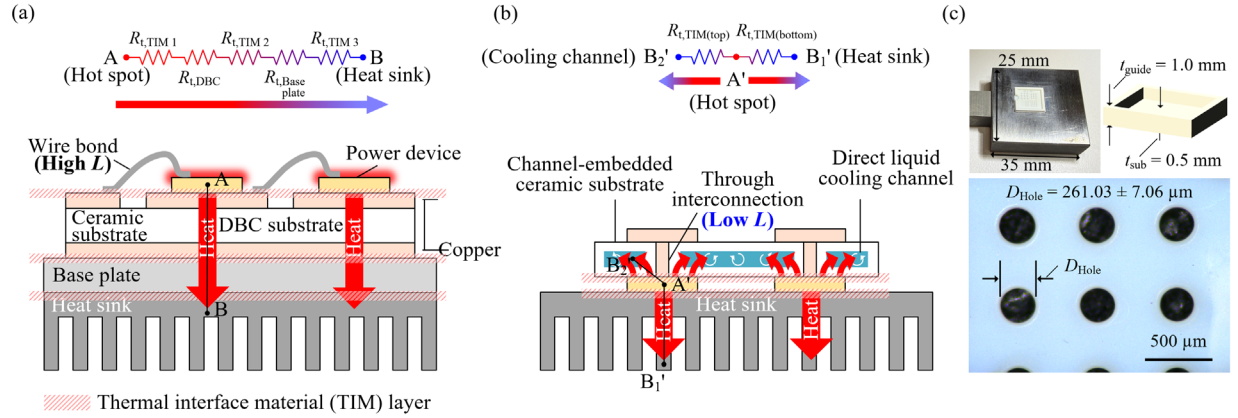


Figure 1. Ceramic substrate for hybrid dual-side cooling power device module. (a) Conventional power device module with wire bonding connection using single-side cooling. (b) Proposed power device module packaging employing 3D-printed channel embedded ceramic through-hole substrate using dual-side cooling with heat sink and direct liquid cooling channel. (c) Optical images of 3D-printed ceramic substrate with through holes for vertical interconnection.